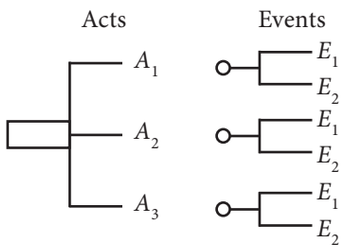
Events (or States of nature): The events identify the occurrences, which are outside of the decision maker' s control and which determine the level of success for a given act. These events are often called ' States of nature' or outcomes. An example of an event or states of nature is the level of market demand for a particular item during a stipulated time period.

A set of states of nature may be represented in any one of the following ways:

$S = \{S_1, S_2, \ldots, S_n\}$ or $E = \{E_1, E_2, \ldots, E_n\}$ or $\Omega = \{\theta_1, \theta_2, \theta_3\}$

For example, if a washing powder is marketed, it may be highly liked by outcomes (outcome θ_1) or it may not appeal at all (outcome θ_2) or it may satisfy only a small fraction, say 25% (outcome θ_3)

$\therefore \Omega = \{\theta_1, \theta_2, \theta_3\}$



In a tree diagram the places are next to acts. We may also get another act on the happening of events as follows:

In a matrix form, they may be represented as either of the two ways.

States of nature ⟶ Acts ↓	S_1	S_2
A_1 A_2		

or

Acts ⟶ States of nature ↓	$A_1 A_2, \ldots A_n$
S_1 S_2	

Pay-off: The result of combinations of an act with each of the states of nature is the outcome and monetary gain or loss of each such outcome is the pay-off. This means that the expression pay-off should be in quantitative form.

Pay -off may be also in terms of cost saving or time saving. In general, if there are k alternatives and n states of nature, there will be $k \times n$ outcomes or pay-offs. These $k \times n$ payoffs can be very conveniently represented in the form of a $k \times n$ pay-off table.

States of nature	Decision alternative			
	A_1	A_2	.....	A_k
E_1	a_{11}	a_{12}	.....	a_{1k}
E_2	a_{21}	a_{22}	.....	a_{2k}
.	.	.	.....	.
.	.	.	.....	.
	.	.	.....	.
E_n	a_{n1}	a_{n2}	.....	a_{nk}

Where a_{ij} = conditional outcome (pay-off) of the ith event when j^{th} alternative is chosen. The above pay-off table is called pay-off matrix.

10.3.2 Situations- Certainty and uncertainty

Types of decision making: Decisions are made based upon the information data available about the occurrence of events as well as the decision situation. There are two types of decision making situations: certainty and uncertainty.

Decision making under certainty: In this case the decision maker has the complete knowledge of consequence of every decision choice with certainty. In this decision model, assumed certainty means that only one possible state of nature exists.

Decision making under uncertainty: Under conditions of uncertainty, only pay-offs are known and nothing is known about the lilkelihood of each state of nature.

XII Std - Business Maths & Stat EM Chapter 10.indd 258 2/29/2020 11:45:21 AM